

EXPERIMENT – 14

PRODUCER–CONSUMER PROBLEM USING SYSTEMC

CODE:

```
#include <systemc.h>

// Producer Module

SC_MODULE(Producer){

    sc_fifo_out<int> out;

    void produce(){

        for (int i = 1; i <= 5; i++){

            cout << "Time: " << sc_time_stamp()

                << " Produced: " << i << endl;

            out.write(i);

            wait(1, SC_SEC);

        }

    }

    SC_CTOR(Producer){

        SC_THREAD(produce);

    }

};

// Consumer Module

SC_MODULE(Consumer){

    sc_fifo_in<int> in;

    void consume(){

        int value;

        for (int i = 1; i <= 5; i++) {
```

```

        in.read(value);

        cout << "Time: " << sc_time_stamp()
              << " Consumed: " << value << endl;

        wait(1, SC_SEC);
    }
}

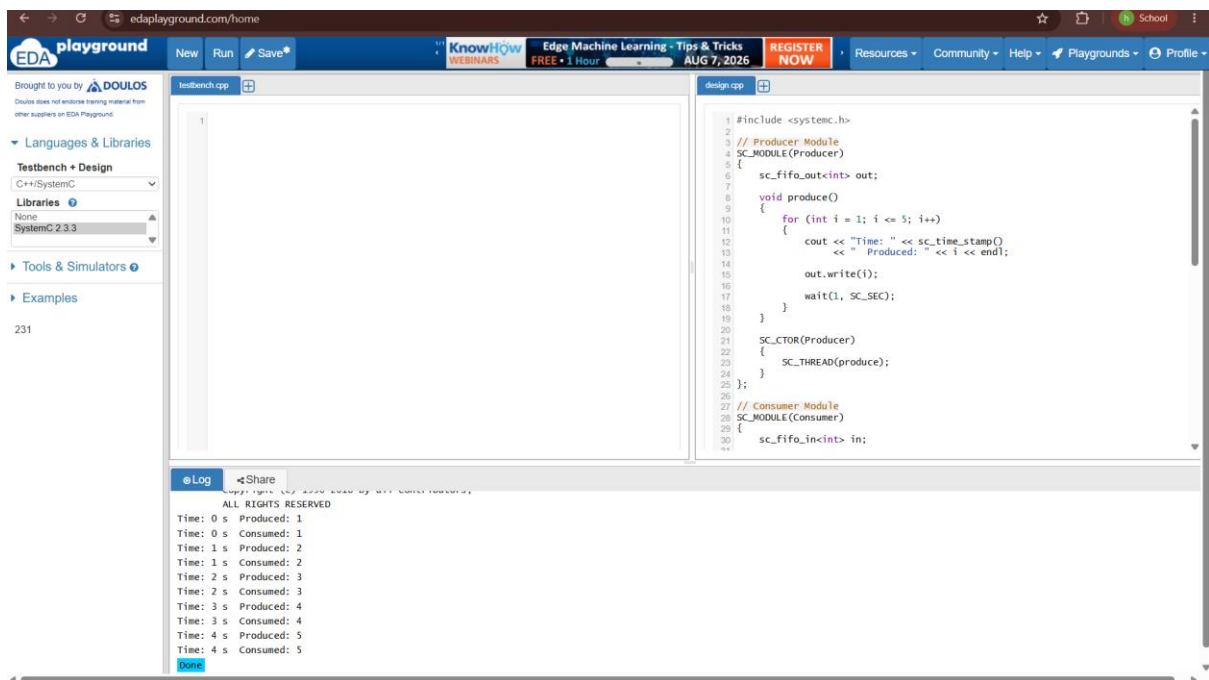
SC_CTOR(Consumer){
    SC_THREAD(consume);
}

};

int sc_main(int argc, char* argv[]){
    sc_fifo<int> fifo(5);
    Producer producer("Producer");
    Consumer consumer("Consumer");
    producer.out(fifo);
    consumer.in(fifo);
    sc_start(10, SC_SEC);
    return 0;
}

```

SIMULATION:



OUTPUT:

